

## 1. Introduction

This guide explains the complete installation procedure for deploying the AI-Powered SOC Assistant integrated with Wazuh SIEM.

The deployment consists of:

- Docker-based Wazuh Server
- Windows Endpoint Agent
- Python AI Monitor
- AI Analyzer

The guide follows the exact implementation performed during the project.

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## 2. System Requirements

### Host Machine

| Component        | Requirement                 |
|------------------|-----------------------------|
| Operating System | Kali Linux                  |
| RAM              | Minimum 8 GB                |
| CPU              | Intel i5 / Ryzen 5 or above |
| Storage          | Minimum 50 GB               |

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### Endpoint Machine

| Component        | Requirement   |
|------------------|---------------|
| Operating System | Windows 11    |
| RAM              | 4 GB or above |

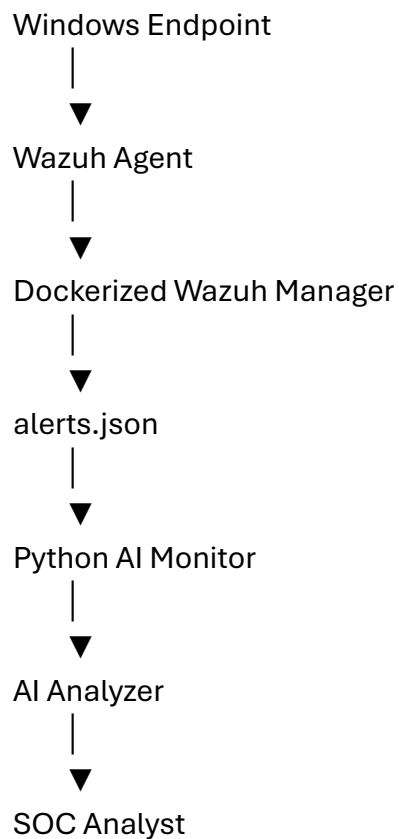
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## 3. Software Requirements

- Docker Desktop / Docker Engine
- Docker Compose
- Git

- Python 3.13
  - Wazuh 4.12
  - Windows Wazuh Agent
  - Visual Studio Code (optional)
- 

#### 4. Project Architecture



#### 5. Docker Installation

##### Step 1

Update packages.

```
sudo apt update
```

---

##### Step 2

Install Docker.

```
sudo apt install docker.io
```

---

### Step 3

Enable Docker.

```
sudo systemctl enable docker
```

---

### Step 4

Start Docker.

```
sudo systemctl start docker
```

---

### Step 5

Verify Docker installation.

```
docker ps
```

Expected Result:

Docker runs successfully without errors.

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## 6. Wazuh Deployment

### Step 1

Clone Wazuh Docker repository.

```
git clone https://github.com/wazuh/wazuh-docker.git
```

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### Step 2

Navigate to the deployment directory.

```
cd wazuh-docker/single-node
```

---

### Step 3

Generate SSL certificates.

```
docker compose -f generate-indexer-certs.yml run --rm generator
```

---

## Step 4

Deploy Wazuh.

```
docker compose up -d
```

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## Step 5

Verify containers.

```
docker ps
```

Expected containers:

- Wazuh Manager
  - Wazuh Dashboard
  - Wazuh Indexer
- 

## 7. Accessing Wazuh Dashboard

Open the browser.

`https://<Manager-IP>`

Example:

`https://192.168.xx.xxx`

Login using administrator credentials.

Verify:

- Dashboard opens successfully.
  - Manager is healthy.
- 

## 8. Windows Agent Installation

Download the Wazuh Agent MSI installer.

Install the agent on the Windows system.

During installation specify:

- Wazuh Manager IP
- Agent Name

Example:

Manager IP:

192.168.xx.xxx

Agent Name:

Balaganesh

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## 9. Agent Registration

Open Command Prompt as Administrator.

Navigate to:

```
cd "C:\Program Files (x86)\ossec-agent"
```

Authenticate the agent.

```
agent-auth.exe -m <Manager-IP>
```

Expected Output:

Valid key received

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Start the service.

```
NET START WazuhSvc
```

Expected Output:

The Wazuh service was started successfully.

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## 10. Agent Verification

Login to Wazuh Dashboard.

Navigate to:

Agents

Verify:

- Agent is Active
  - Agent Status = Connected
-

## 11. Python Environment Setup

Create project directory.

```
mkdir AI_Wazuh_Lab
```

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Navigate to project.

```
cd AI_Wazuh_Lab
```

---

Create virtual environment.

```
python3 -m venv venv
```

---

Activate environment.

```
source venv/bin/activate
```

---

Install dependencies.

```
pip install openai python-dotenv
```

---

Verify packages.

```
pip list
```

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## 12. AI Module Setup

Create the following files.

```
ai_monitor.py
```

```
ai_analyzer.py
```

```
config.py
```

```
.env
```

Store the OpenAI API key (if using OpenAI) or configure the local AI analyzer according to your implementation.

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### 13. Running the AI Monitor

Execute:

```
python ai_monitor.py
```

Expected Output:

```
===== Latest Wazuh Alert =====
```

Agent:

Balaganesh

Rule ID:

60106

Threat Category:

Authentication

Risk Level:

Low

MITRE:

T1078

Recommendations

Review activity

Continue monitoring

Confidence:

92%

The monitor should continue running and automatically process new alerts as they are generated.

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### 14. Testing

Generate Windows events:

- Successful Login

- Failed Login
- Logoff
- Account Lock
- PowerShell Execution (optional)

Verify:

- Alert appears in Wazuh Dashboard.
  - AI monitor detects the alert.
  - AI analysis is displayed.
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## **15. Troubleshooting**

### **Docker Containers Not Running**

Check:

`docker ps`

Restart:

`docker compose restart`

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### **Dashboard Not Opening**

Verify:

- Docker containers are running.
  - Correct Manager IP is used.
  - Browser certificate warning is accepted (if using self-signed certificates).
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### **Windows Agent Not Connected**

Verify:

`NET START WazuhSvc`

Re-authenticate:

`agent-auth.exe -m <Manager-IP>`

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## **AI Monitor Shows JSON Error**

Cause:

Incomplete alert written to alerts.json.

Solution:

Ignore partial entries and continue monitoring.

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## **OpenAI API Error**

Possible causes:

- Invalid API key (401)
- Insufficient quota (429)

Solution:

Verify API configuration or use the local AI analyzer until API access is available.